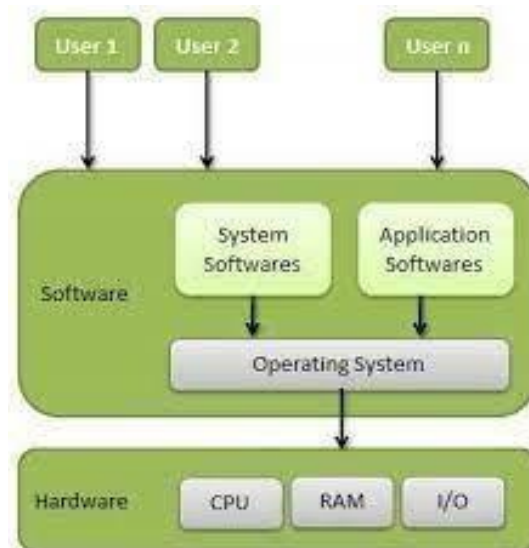


Unit 1

Overview

What is Operating System? What does operating system do?

- A program that acts as an interface between a user of a computer and the computer hardware



- Operating system goals:
 - Execute user programs and make solving user problems easier
 - Make the computer system convenient to use
 - Use the computer hardware in an efficient manner

Functions of Operating System:

1. Memory Management
2. Processor Management
3. Device Management
4. File Management
5. Security
6. Multitasking system
7. Error Detection
8. Co-ordination between other software's and users
9. Work management
10. Resource Management
11. Input Output Control

1. **Memory Management** - Memory management is **the process of controlling and coordinating a computer's main memory**. As all the files, programs, processes etc. are stored in secondary memory. In order to execute these files, they have to be loaded in main memory because the CPU does the execution and takes the files from main memory only. OS ensures that blocks of memory space are properly managed and allocated so the operating system (OS), applications and other running processes have the memory they need to carry out their operations & executions. So allocation and de-allocation of the main memory is done by OS

2. **Processor Management**- Process management involves the creation, scheduling, termination of the process by the operating system.

Ex: If in my Laptop I'm working on MS-word, then opened Windows media player, also along with that I have opened gaming app. This means I have opened multi processes.

And Operating System executes all these processes. It is done by Process scheduling. Scheduling -is a set of policies and mechanisms which controls the order in which the work to be done is completed.

OS decides which process should be executed first and how much time it requires. So OS allocates process to the processor.

3. **Device Management**- Operating System maintains the status of all the devices connected to the system. Allocation of the device to the process is also handled by OS.
4. **File Management** - The file system is the component of the operating system that manages files. It provides a means of storing data and gaining access to the contents of files, including data and applications. Also an interface is provided by the operating system for creating file system backups. OS also tracks the status of all files and where the information is stored.
5. **Security** - OS security refers to specified steps or measures used to protect the OS from threats, viruses, worms, malware or remote hacker intrusions

1. Authentication: Authentication is one of the safeguards employed by operating systems to guarantee that the person accessing a program is permitted and valid.

2. Backup and Restore: The operating system includes software components that enable users to create backups or copies of data, as well as restore these backups when necessary.

6. **Multitasking System** - Multitasking is now a standard feature of current operating systems. They are capable of performing numerous jobs at once. For example, if a user is working in Excel and wants to print a page, he does not need to interrupt his work. With just one command, the page will be printed, and the user can access the drives

without having to close the tab. As a result, multitasking is one of the best characteristics of operating systems.

7. **Error Handling-**

- In the operating system, errors can occur at any time and in any location.
- When errors are occurred, debugging is done by OS to avoid the malfunctioning of the system. Sometimes OS sends reports to the user to rectify the errors occurred.
- A bug in the CPU, I/O devices, or memory hardware can all cause problems.
- The Operating System is continually looking for problems such as power failure, connection failure , lack of paper in printers etc.
- In the worst-case situation, the application's error handling procedures compel the user to log out and the system to shut down.

8. **Co-ordination between software & hardware** – OS provides all language translators such as compilers, interpreters, assemblers to user for smooth condition of the process.

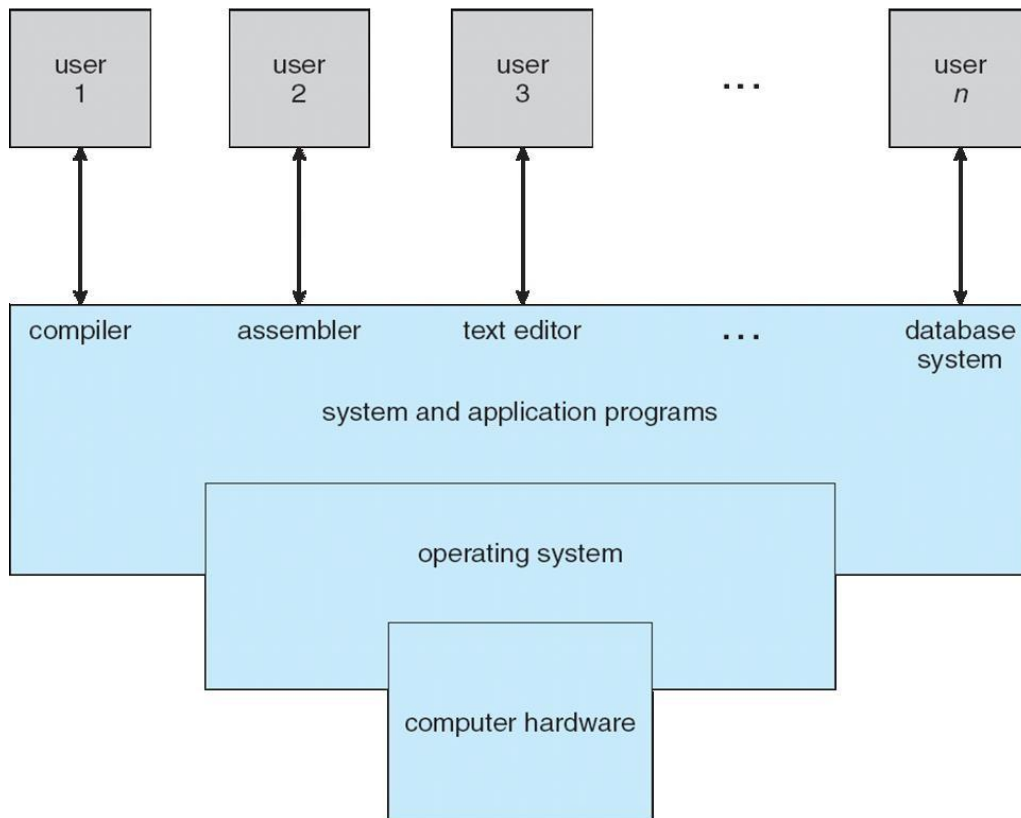
9. **Work Management** - OS determines the order in which the activities are carried out, with a queue indicating the jobs that await execution. Work management is the key to controlling all tasks and their required resources without any conflicts. It helps streamline processes to better schedule tasks and effectively manage resources without compromising performance.

10. **Resource Management** -The hardware and software of a computer are closely interconnected. Each of the two does not work without the other. As a result, the operating system balances these two resources. OS examines all tasks systematically and verifies if all computer components are operational

11. **Input and Output Control-** The three main jobs of a computer are Input, Output, and Processing. The operating system also controls a computer's input and output activities. It creates a dictionary of input and output tasks that can run anytime the user issues a command. The operating system has a set of programs that allow it to handle interrupts and process the input and output instructions provided by the user.

Computer System Organization

Four Components of a Computer System



Computer system can be divided into four components:

1. Hardware – provides basic computing resources.
Ex: CPU, memory, I/O devices
2. Operating system- Controls and coordinates use of hardware among various applications and users.
Ex: Microsoft Windows, Linux, macOS etc
3. Application programs – define the ways in which the system resources are used to solve the computing problems of the users
Ex: Word processors, compilers, web browsers, database systems, video games
4. Users
Ex: People, machines, other computers

Note: (System software examples include operating systems, Compiler, and Assembler. Application software examples include Google Chrome, VLC media player, Microsoft Word, and Photoshop)

Important Terms

1. Operating System:

A program that acts as an intermediary between a user of a computer and the computer hardware.

Operating system goals:

- Execute user programs and make solving user problems easier
- Make the computer system convenient to use
- Use the computer hardware in an efficient manner

2. Bootstrap program:

- The initial program that runs when a computer is powered up or rebooted
- It is stored in the ROM
- It must know how to load the OS and start executing that system.
- It must locate and load into memory the OS kernel.

3. System Calls:

In computing, a system call is the programmatic way in which a computer program requests a service from the kernel of the operating system. System call is the only entry point for the kernel system. These services are offered with the help of an API (Application Program Interface).

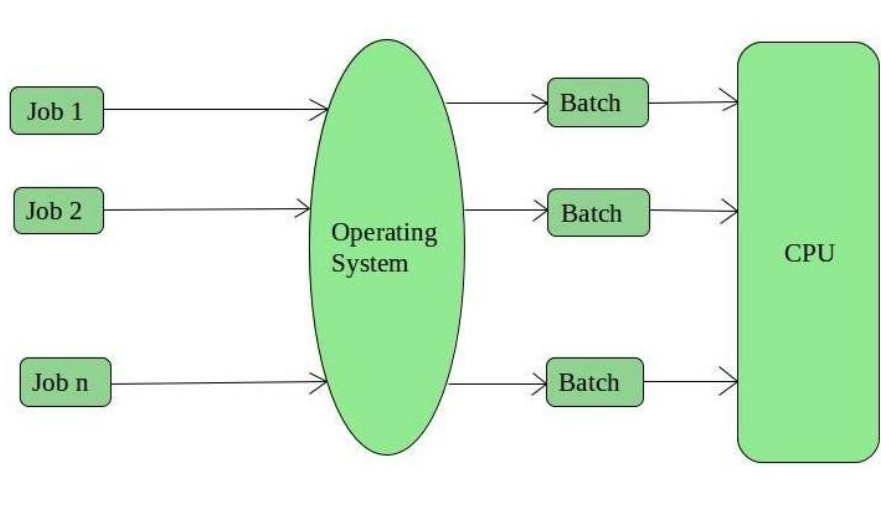
4. Kernel:

The kernel is the heart of the operating system. When the system boots up, the kernel is the first program that is loaded in memory after the bootloader because the kernel handles the rest of the functions of the system for the OS. The kernel remains in the memory until the OS shuts down.

Types of Operating System.

1. Batch Operating System
2. Multiprogrammed Operating System
3. Multitasking Operating System
4. Real Time OS
5. Distributed
6. Clustered
7. Embedded OS

1. Batch Operating System



The users of a batch operating system do not interact with the computer directly. Each user prepares his job on an off-line device like punch cards and submits it to the computer operator. To speed up processing, jobs with similar needs are batched together and run as a group. The programmers leave their programs with the operator and the operator then sorts the programs with similar requirements into batches.

This grouping of batches is done with the help of the operator. When we have the batches ready, the execution is done one by one for the batches which means batch-wise.

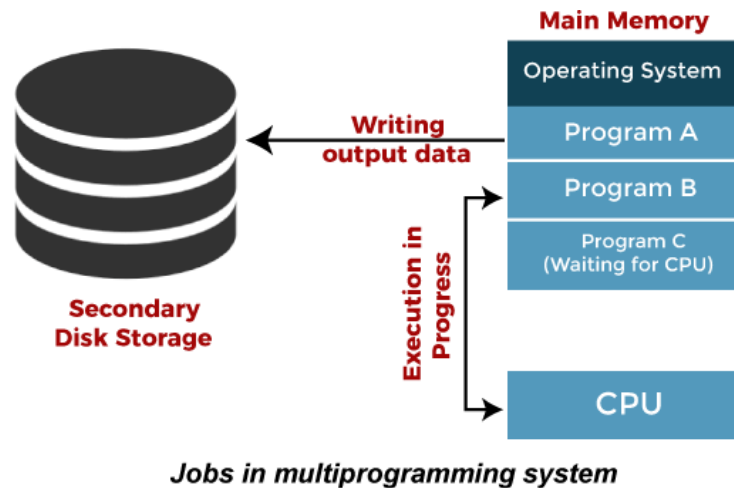
The problems with Batch Systems are as follows –

- Lack of interaction between the user and the job.
- CPU is often idle, because the speed of the mechanical I/O devices is slower than the CPU.
- If a job fails, the other jobs will have to wait for an unknown time.
- Batch systems are sometimes costly.
- Difficult to debug.

2. Multiprogramming Operating System

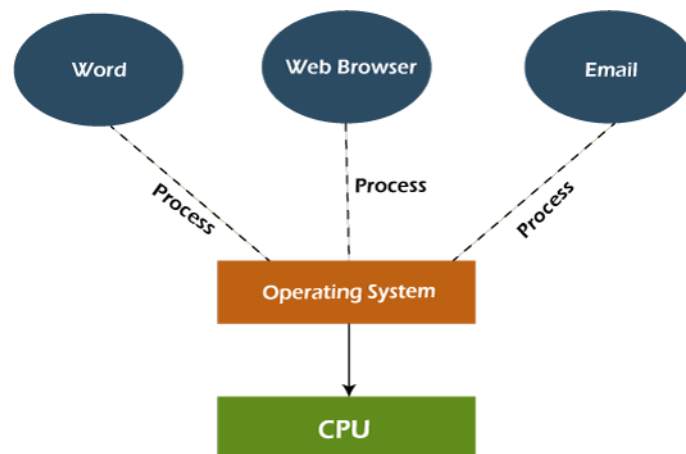
Multiprogramming is an extension to batch processing where the CPU is always kept busy. Each process needs two types of system time: CPU time and IO time.

- In a multiprogramming environment, when a process does its I/O, The CPU can start the execution of other processes. Therefore, multiprogramming improves the efficiency of the system.



3. Time-Sharing Operating Systems –

Each task is given some time to execute so that all the tasks work smoothly. Each user gets the time of CPU as they use a single system. These systems are also known as Multitasking Systems. The task can be from a single user or different users also. After this time interval is over OS switches over to the next task. The multitasking operating system is a logical extension of a multiprogramming system that enables **multiple** programs simultaneously. It allows a user to perform more than one computer task at the same time.

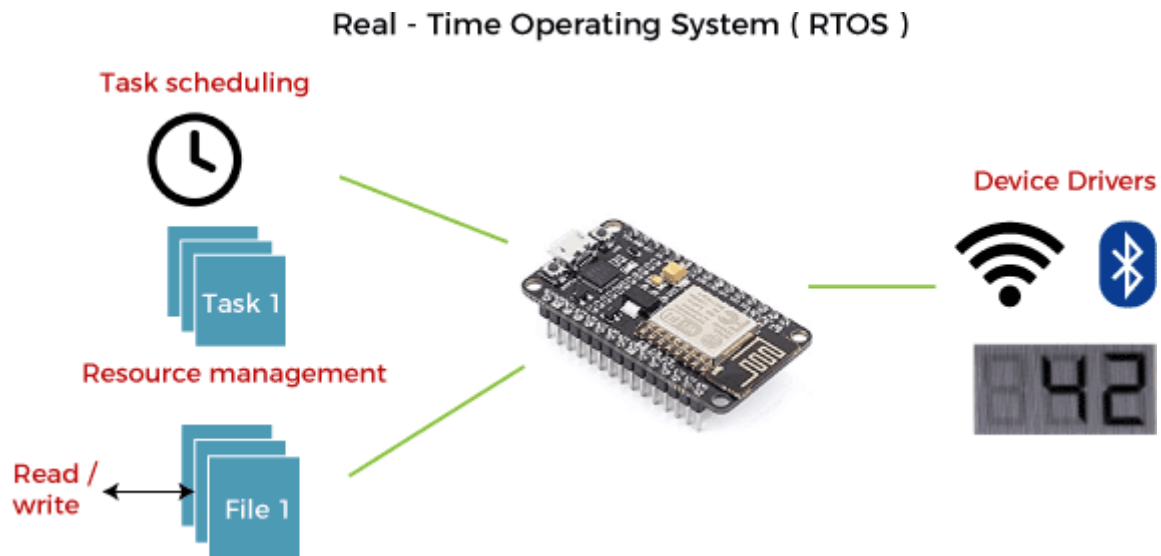


Advantages of Time-Sharing OS:

- Each task gets an equal opportunity
- The time-sharing operating system provides effective utilization and sharing of resources
- CPU idle time can be reduced

4. Real Time Operating System

In Real-Time Systems, each job carries a certain deadline within which the job is supposed to be completed, otherwise, the huge loss will be there, or even if the result is produced, it will be completely useless.



The Application of a Real-Time system exists in the case of military applications, if you want to drop a missile, then the missile is supposed to be dropped with a certain precision.

5. Distributed Operating System

It is a collection of independent computers, interconnected via network capable of communicating and cooperating with each other through a LAN / WAN.

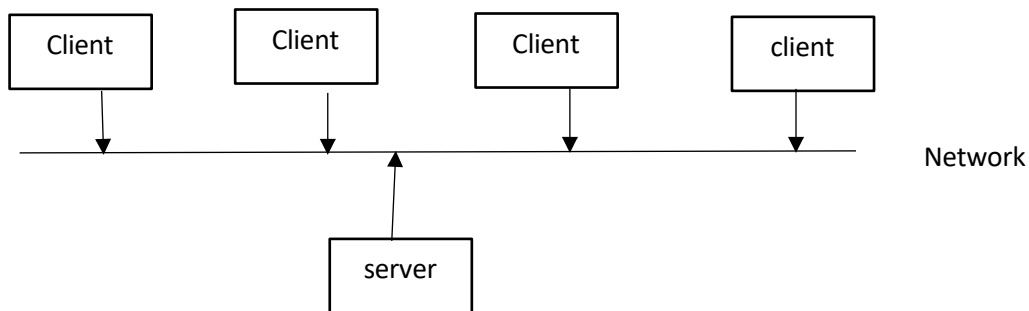
Ex: Telephone network, cellular network

Computer network such as internet

ATM machine

Mobile computing etc.

The above examples follow client- server systems



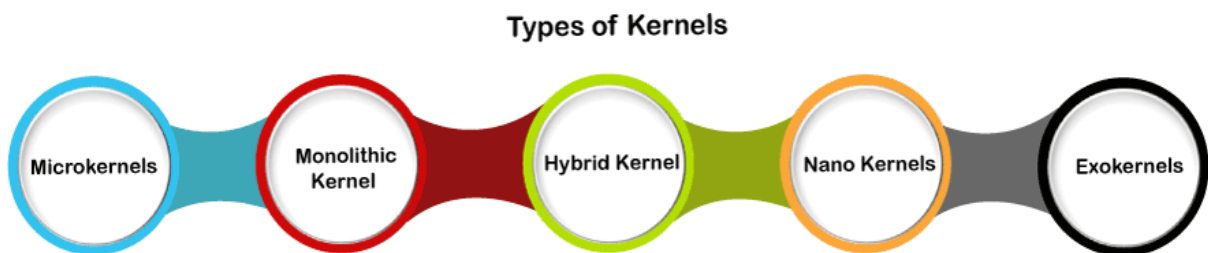
The client sends requests to the server and the server sends the response. A distributed system is loosely coupled as there is no shared memory. Because each system has its own memory and processor.

Advantages of Distributed system

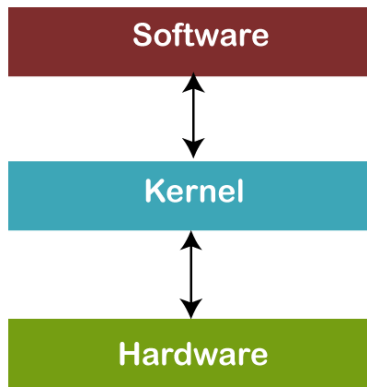
- Resource sharing
- Computation speed up
- Reliability

Types of Kernel

There are mainly five types of Kernel, which are given below:



1. Monolithic Kernels



In a monolithic kernel, the same memory space is used to implement user services and kernel services.

It means, in this type of kernel, there is no different memory used for user services and kernel services.

As it uses the same memory space, the size of the kernel increases, increasing the overall size of the OS.

The execution of processes is also faster than other kernel types as it does not use separate user and kernel space. Linux operating system is monolithic kernel. Android is monolithic kernel because android is modified version of linux OS, google chrome is monolithic kernel.

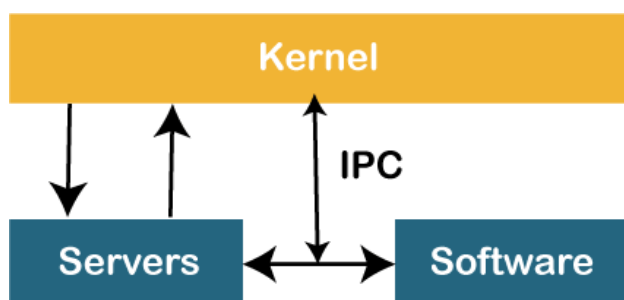
Advantages:

- The execution of processes is also faster as there is no separate user space and kernel space and less software involved.
- It provides CPU scheduling, memory scheduling, file management through System calls only.

Disadvantages:

- If any service generates any error, it may crash down the whole system.
- Large in size and hence become difficult to manage.
- To add a new service, the complete operating system needs to be modified.

2. Microkernel



Microkernel is totally opposite to 1st one , It also provides support to advanced features as compared to microkernel.

Ex: if your working on OS which is based on Micro kernel may be it will demand for the device drivers if you use additional devices with computer.

If you are on Monolithic kernel it will provide support to the additional device for which micro was demanding additional drivers.

it uses different spaces for both the services, so, the size of the microkernel is decreased, and which also reduces the size of the OS.

Ex: we find it in small devices like smart watches are based on microkernel.

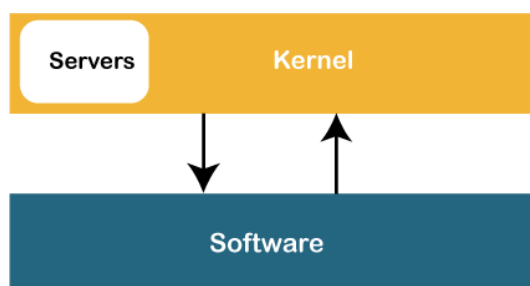
Advantages

- A new service can be easily added without modifying the whole OS.
- In a microkernel, if a kernel process crashes, it is still possible to prevent the whole system from crashing.

Disadvantages

- Since we are using User Space and Kernel Space separately, so the communication between these can reduce the overall execution time.
- Process management is very complicated.
- The messaging bugs are difficult to fix.

3. Hybrid Kernel



Hybrid kernels are also known as **modular kernels**, and it is the combination of both Monolithic and Microkernels. It takes advantage of the speed of monolithic kernels and the modularity of microkernels.

A hybrid kernel can be understood as the extended version of a microkernel with additional properties of a monolithic kernel. These kernels are widely used in commercial OS, such as different versions of MS Windows.

It is much similar to a microkernel, but it also includes some additional code in kernel space to enhance the performance of the system.

Ex of hybrid kernel is windows which is neither micro nor monolithic kernel

Advantages:

- There is no requirement for a reboot for testing.
- Third-party technology can be integrated rapidly.

Disadvantages:

- There is a possibility of more bugs with more interfaces to pass through.
- It can be a confusing task to maintain the modules for some administrators, especially when dealing with issues such as symbol differences.

4. Nanokernel

As the name suggests, in Nanokernel, the complete code of the kernel is very small, which means the code executing in the privileged mode of the hardware is very small. Here the term nano defines a kernel that supports a nanosecond clock resolution.

Examples of Nanokernel **are EROS etc.**

Advantages

- It provides hardware abstractions even with a very small size.

Disadvantages

- Nanokernel lacks system services.

5. Exokernel

Exokernel is still developing and is the experimental approach for designing OS.

This type of kernel is different from other kernels as in this; resource protection is kept separated from management, which allows us to perform application-specific customization.

Advantages:

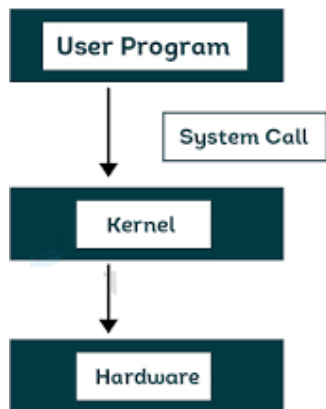
- The exokernel-based system can incorporate multiple library operating systems. Each library exports a different API, such as one can be used for high-level UI development, and the other can be used for real-time control.

Disadvantages:

- The design of the exokernel is very complex.

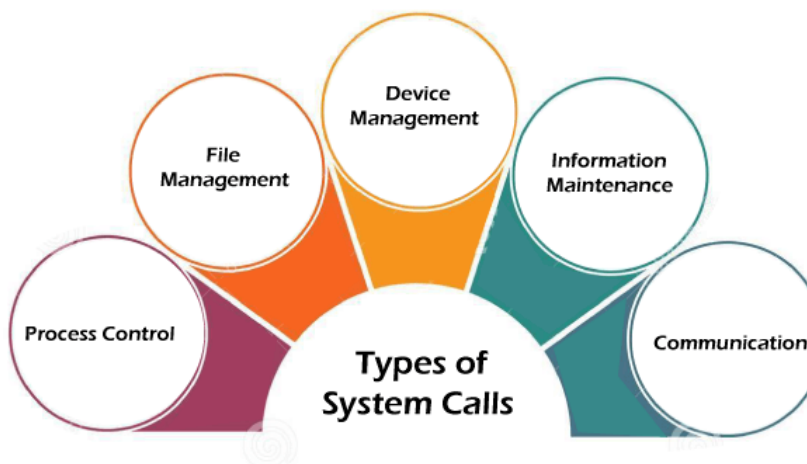
System Calls

System call acts as an interface between software applications and kernel.



In computing, a system call is the programmatic way in which a computer program requests a service from the kernel of the operating system it is executed on. A system call is a way for programs to interact with the operating system. A computer program makes a system call when it makes a request to the operating system's kernel. system calls are available as assembly language instructions.

Types of System Calls



1. **Process Control**
2. **File Management**
3. **Device Management**
4. **Information Maintenance**
5. **Communication**

1) Process Control

Process control is the system call that is used to direct the processes. Some process control examples include creating, load, abort, end, execute, process, terminate the process, etc.

System calls used to control a process provides the following services.

- End the process normally, to abort the process forcefully.
- Load a process to the main memory, and execute the process.
- Create a new process, and terminate the current process.
- Wait for a process to finish its execution. Wait till a certain event occurs, signal when the event has occurred.
- Allocate memory to a process, if the process terminates then free the memory.

2) File Management

This kind of system call provides the following services:

- Create and delete files.
- Open and close the file.
- Read from a file, write to a particular file.
- To get the attribute of a file and to set the attribute of a file.

3) Device Management

A system call that manages the I/O devices provides the following services:

- A process while executing may require devices. Such as access to I/O devices, files system, main memory etc.
- So, it can request a device and after the task is over it can release the device.
- Once the process gets access to a requested device. It can perform read, write or reposition operations.
- The call can be made to attach a device to the executing processor to detach it.

This means logically attaching and detaching the device. Ex when we insert pen drive OS understands that some device has been plugged in. then to detach the device we right click on the icon and click eject pen drive. Though it is physically logged in but logically it is removed.

4) Information Maintenance

The system call that is used to maintain the information of the system ,they fall under this category of system calls.

So, the services provided by this kind of system call are:

- Get the date or time of the system. Set the date or time of the system.
- Get the information about the system. Set the system information.

5) Communications

This kind of system call helps in connecting the system to a network. Services provided by these system calls are:

- Create a new connection to transfer the data. Delete a connection after the transmission gets complete.
- Send a message on a particular connection. Receive a message from a particular connection.
- Attach a particular remote device to the network. Detach a particular remote device from the network.

Kernel

Kernel loads first in memory after OS has been loaded and remains in memory until OS is shutdown.

- As discussed above, Kernel is the core part of an OS(Operating system); hence it has full control over everything in the system. Each operation of hardware and software is managed and administrated by the kernel.
- It is the part of the OS that always resides in computer memory and enables the communication between software and hardware components.
- It is the computer program that first loaded on start-up the system (After the bootloader).

Operating System Services

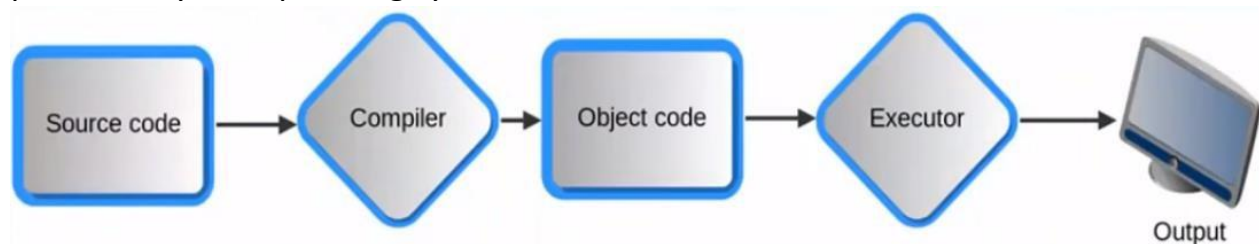
Following are a few common services provided by an operating system –

1. Program execution
2. I/O operations
3. File System manipulation
4. Communication
5. Error Detection
6. Resource Allocation
7. Protection
8. Accounting

We have 2 services from OS that is user services and system services.

2) Program Execution

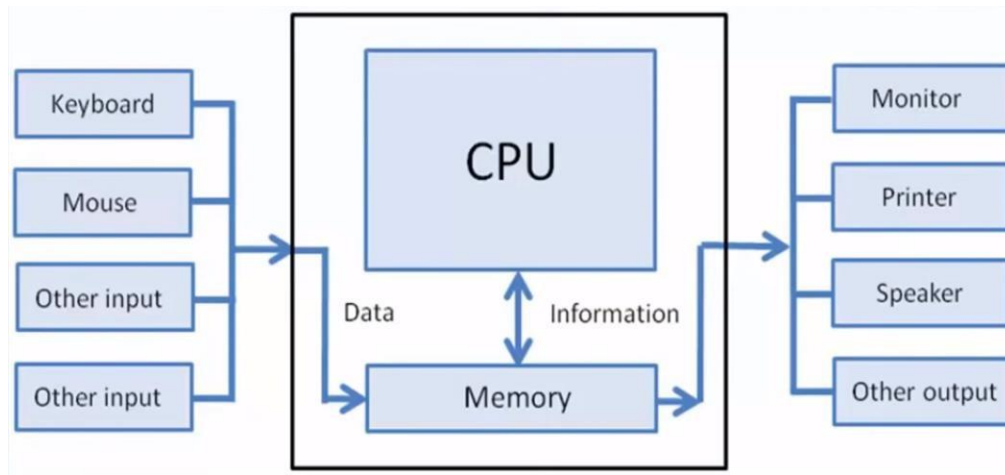
The system will load the program into memory and run the program. This is the service provided by the operating system.



3) I/O operations

A running program may require I/O which involve files or I/O devices. This input/output operations should be provided by the operating system. A user cannot directly control input/output devices. Now we may feel that mouse and keyboard are under our control. But operating system is between user and the system that controls the usage of

I/O devices.



4) File system manipulation



File System Manipulation

File system manipulation or file system management involves files. There are many files and directories in our system that has to be used. Now the operating system must control how these files are manipulated or managed. Sometimes we may have to create, delete, modify or search for a given file. So all this is controlled by OS. It also controls the permission that is given to certain programs or users for the access of certain files. So all the files cannot be allowed to be used by every program or by every user. There is access restrictions. This access restriction is controlled by OS.

5) Communication.



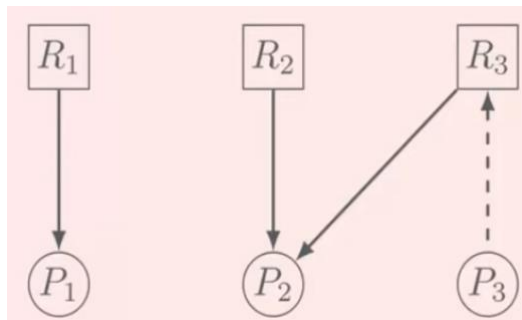
The operating system manages the exchange of data and programs among different computers connected over a network. This communication is accomplished using message passing and shared memory. Communication between processes. A problem that is in execution is known as process.

6) Error Detection



Errors can always occur in the course of computing. And operating system needs to be constantly aware of possible errors that can occur. So there are many kinds of errors that can occur. Like there are errors that can occur in the CPU, memory, hardware or errors in input/output devices. Example as in the above figure, we have a printer and we are printing something and suddenly the paper runs out. Now the printer is not able to print because there is no more available paper. Now this is an error and operating system must be able to handle this error. So that it ensures correct and consistent computing. The OS must be able to handle errors and still carried on even if errors occur.

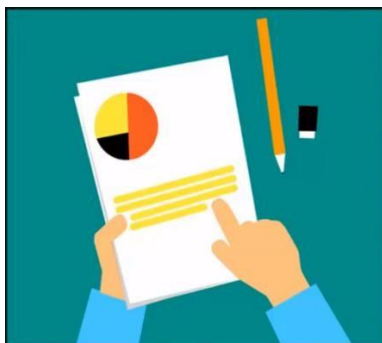
7) Resource Allocation



Allocating resources to different processes to different users. Resources may be CPU, files, Input/output devices, main memory and so

There should never be a scenario where a process is waiting for resource and it never gets it. And there should never be a scenario where a resource is held by a particular process and it is never released.

8) ACCOUNTING



Accounting is keeping track of which user use how much and what kind of resources. This record keeping may be used for accounting or simply accumulating usage statistics. This accounting can be a valuable tool for researchers who wish to reconfigure the system or improve computing services.

9) Protection and Security



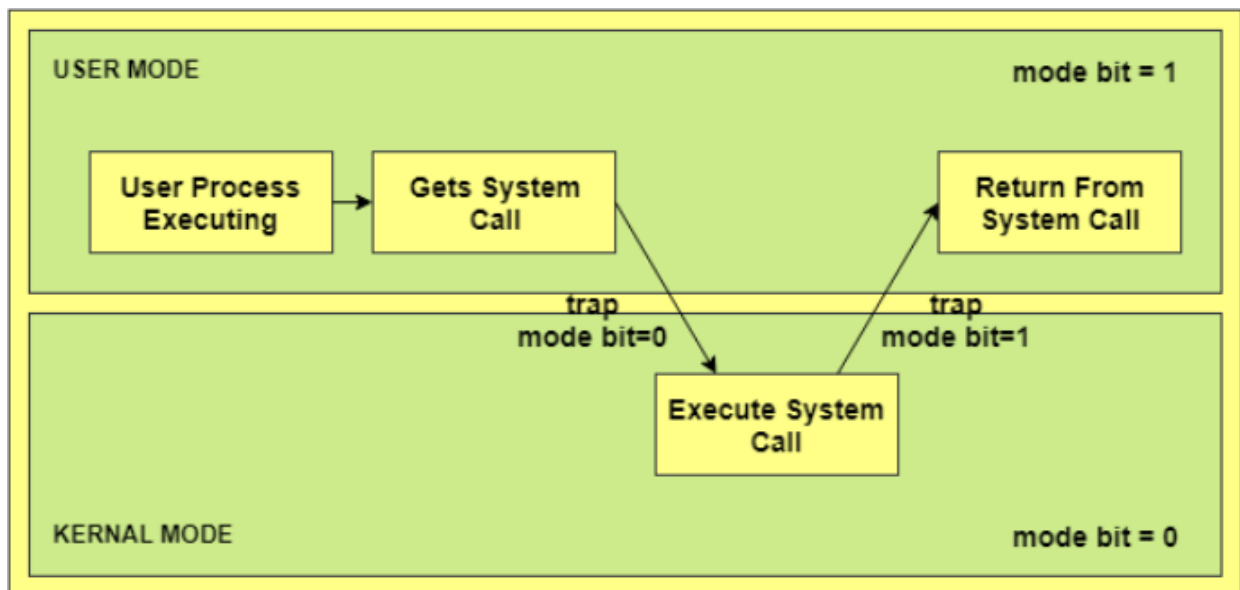
Our data should be secure and everything that we do must be protected. When several different processes are executing at the same time. It should not be possible for one process to interfere with other process. Ex email accounts or social media accounts should be protected with passwords.

Operating-System Operations

We need two separate modes of operation: **user mode** and **kernel mode** (also called supervisor mode, system mode, or privileged mode).

1. Dual- Mode Operation
2. Timer

1. Dual- Mode Operation



All operating systems are interrupt driven. A trap is a special type of interrupt. A trap is a software generated. A trap is generated in 2 cases. Whenever there is error then software generates a trap. Other is when it requires services from Operating System.

Mode bit is used to differentiate between user mode and kernel mode.

Mode bit =0----- Kernel mode

Mode bit =1 -----User mode

- In order to ensure the proper execution of the operating system, we must be able to distinguish between the execution of operating-system code and user defined code.

- A bit, called the **mode bit**, is added to the hardware of the computer to indicate the current mode: kernel (0) or user (1).
- With the mode bit, we are able to distinguish between a task that is executed on behalf of the operating system and one that is executed on behalf of the user.
- When the computer system is executing on behalf of a user application, the system is in user mode.
- However, when a user application requests a service from the operating system (via a system call), it must transition from user to kernel mode to fulfil the request.
- At system boot time, the hardware starts in kernel mode.
- The operating system is then loaded and starts user applications in user mode.
- Whenever a trap or interrupt occurs, the hardware switches from user mode to kernel mode (that is, changes the state of the mode bit to 0).
- Thus, whenever the operating system gains control of the computer, it is in kernel mode.
- The system always switches to user mode (by setting the mode bit to 1) before passing control to a user program.
- The dual mode of operation provides us with the means for protecting the operating system from errant users
- Another example is banks example where the bank people are kernel mode and we are user mode. Another ex is when we write a program to add $2+2=4$.To display on monitor we go to kernel mode to access to monitor that is hw and then shift to user mode when it is displayed on monitor.
- If there was no mode bit and therefore no dual mode, a user program running awry can wipe out the operating system by writing over it with data; and multiple programs are able to write to a device at the same time, with possibly disastrous results.

User Mode vs Kernel Mode

User Mode

The system is in user mode when the operating system is running a user application such as handling a text editor. The transition from user mode to kernel mode occurs when the application requests the help of operating system or an interrupt or a system call occurs.

The mode bit is set to 1 in the user mode. It is changed from 1 to 0 when switching from user mode to kernel mode.

Kernel Mode

The system starts in kernel mode when it boots and after the operating system is loaded, it executes applications in user mode. There are some privileged instructions that can only be executed in kernel mode.

These are interrupt instructions, input output management etc. If the privileged instructions are executed in user mode, it is illegal and a trap is generated.

The mode bit is set to 0 in the kernel mode. It is changed from 0 to 1 when switching from kernel mode to user mode.

2. Timer

- We must ensure that the operating system maintains control over the CPU. We must prevent a user program from getting stuck in an infinite loop or not calling system services and never returning control to the operating system. (Ex: if you have opened an application like paint or text editor and nothing is working then it is infinite loop. Means it has opened but no operations happening this is stuck in infinite loop. In this situation system call services also fails).
- Here operating system tries to break this infinite loop by setting a timer
- To accomplish this goal, we can use a timer. A timer can be set to interrupt the computer after a specified period. The period may be fixed (for example, 1/60 second) or variable (for example, from 1 millisecond to 1 second). A variable timer is generally implemented by a fixed-rate clock and a counter.
- If the timer interrupts, control transfers automatically to operating system .
- The operating system sets the counter. Every time the clock ticks, the counter is decremented. When the counter reaches 0, an interrupt occurs. For instance, a 10-bit counter with a 1-millisecond clock allows interrupts at intervals from 1 millisecond to 1,024 milliseconds, in steps of 1 millisecond.
- Before turning over control to the user, the operating system ensures that the timer is set to interrupt. If the timer interrupts, control transfers automatically to the operating system, which may treat the interrupt as a fatal error or may give the program more time.
- Thus, we can use the timer to prevent a user program from running too long. A simple technique is to initialize a counter with the amount of time that a program is allowed to run. A program with a 7-minute time limit, for example, would have its counter initialized to 420.
- Every second, the timer interrupts and the counter is decremented by 1. As long as the counter is positive, control is returned to the user program. When the counter becomes negative, the operating system terminates the program for exceeding the assigned time limit.
- Ex: while working in windows of photoshop and suddenly it gets stucked. After timer the OS takes control and then we get a message application(photoshop) not working, wait for program to respond or close the program. When we click wait the program to respond again a timer is set.

Introduction to Linux and Basic Linux Commands

1. Introduction to Linux

Linux is an open-source operating system that manages computer hardware and software resources. It provides an interface between the user and the computer system.

Linux is widely used in:

1. Servers
2. Supercomputers
3. Cloud computing
4. Embedded systems
5. Cybersecurity applications
6. Features of Linux:
7. Open Source: The source code is freely available and can be modified.
8. Multi-user: Multiple users can work on the system at the same time.
9. Multitasking: It can run multiple programs simultaneously.
10. Security: Provides strong user authentication and permission control.
11. Command Line Interface: Users can interact with the system using commands.

2. Linux Distributions

A Linux Distribution (Distro) is a complete operating system based on the Linux kernel along with additional software and tools.

Some popular Linux distributions are:

1. Ubuntu

User-friendly Linux distribution.

Commonly used by beginners and developers.

2. Fedora

Developed by the Fedora Project.

Used for testing new technologies.

3. Debian

Known for stability and reliability.

Used in many servers.

4. Red Hat Enterprise Linux (RHEL)

Used mainly in enterprise and commercial environments.

5. Kali Linux

Used for cybersecurity, ethical hacking, and penetration testing.

3. Basic Linux Commands

Linux commands are instructions given to the operating system through the terminal to perform different tasks.

1. ls Command (List)

Purpose: Displays files and directories present in the current location.

Syntax: ls

Example: ls

Output: Documents Downloads file.txt

2. pwd Command (Print Working Directory)

Purpose: Shows the current directory path where the user is working.

Syntax: pwd

Example Output: /home/student

3. cp Command (Copy)

Purpose: Used to copy files or directories from one location to another.

Syntax: cp source destination

Example: cp file1.txt file2.txt

This creates a copy of file1.txt with the name file2.txt.

4. mv Command (Move)

Purpose: Used to move files or rename files.

Syntax: mv source destination

Example: mv old.txt new.txt

This renames old.txt as new.txt.

5. rm Command (Remove)

Purpose: Used to delete files.

Syntax: rm filename

Example: rm sample.txt

This removes the file permanently.

6. man Command (Manual)

Purpose: Displays the help manual for any Linux command.

Syntax: `man command_name`

Example: `man ls`

It shows detailed information about the `ls` command.

4. Command Line Usage

The Command Line Interface (CLI) allows users to interact with the computer by typing commands instead of using graphical interfaces.

Advantages of Command Line:

- Faster execution of tasks.
- Requires less system memory.
- Useful for system administration.
- Provides more control over the operating system.

Example:

`mkdir project`

This command creates a new directory named "project".

5. System Call Tracing

A system call is a request made by a program to the operating system for performing a specific task.

Examples of system calls:

- Opening a file
- Reading data
- Creating a process
- Allocating memory
- System Call Tracing:

System call tracing is the process of monitoring the system calls made by a program during execution.

A Linux tool called `strace` is used for tracing system calls.

Example: `strace ls`

This displays the system calls used when executing the ls command.

Uses of System Call Tracing:

- Debugging programs
- Understanding program execution
- Finding errors in applications
- Monitoring system activities

6. Mode Switching Scenarios in Linux

Linux operates in two important modes:

1. User Mode

Normal programs run in user mode.

Users have limited access to hardware resources.

Provides security by preventing direct hardware access.

Example:

Running a text editor

Browsing files

2. Kernel Mode

The Linux kernel runs in kernel mode.

It has complete access to hardware and system resources.

Example:

Managing memory

Controlling devices

Mode Switching:

When a user program needs a service from the operating system, it switches from user mode to kernel mode using a system call.

Example:

User opens a file.

Program requests file access using a system call.

CPU switches to kernel mode.

Kernel performs the operation.

Control returns to user mode.

7. Command Output Prediction

Command output prediction means understanding what result will be displayed after executing a Linux command.

Example:

Command: pwd

Output: /home/user

Explanation: The command displays the current working directory.

Example:

Command: ls

Output: file1.txt

Explanation: Displays all files available in the current directory.

Importance of Command Prediction:

- Helps in debugging errors.
- Improves command-line skills.
- Allows users to understand system behavior.